

Numerical Questions of iT project management

Question 1: Progress vs. Planned Performance

A project has the following planned progress data for the first 5 months:

- Month 1: 10% complete
- Month 2: 30% complete
- Month 3: 55% complete
- Month 4: 75% complete
- Month 5: 100% complete

The actual progress of the project is as follows:

- Month 1: 15% complete
- Month 2: 25% complete
- Month 3: 50% complete
- Month 4: 70% complete
- Month 5: 90% complete

Question:

- (a) Draw the S Curve for the planned vs. actual progress of the project.
- (b) Analyze the gap between the planned and actual progress. Which months show a delay in the project?

Solution:

Step 1: Tabular Representation of Data

Month	Planned Progress (%)	Actual Progress (%)	Difference (Actual - Planned)
1	10%	15%	+5%
2	30%	25%	-5%
3	55%	50%	-5%
4	75%	70%	-5%
5	100%	90%	-10%

Step 2: Drawing the S-Curve

Explanation:

1. **Planned Progress** and **Actual Progress** data need to be plotted on a graph (S-curve).
2. **X-axis:** Represents months (1 to 5).
3. **Y-axis:** Represents cumulative progress in percentage (0% to 100%).

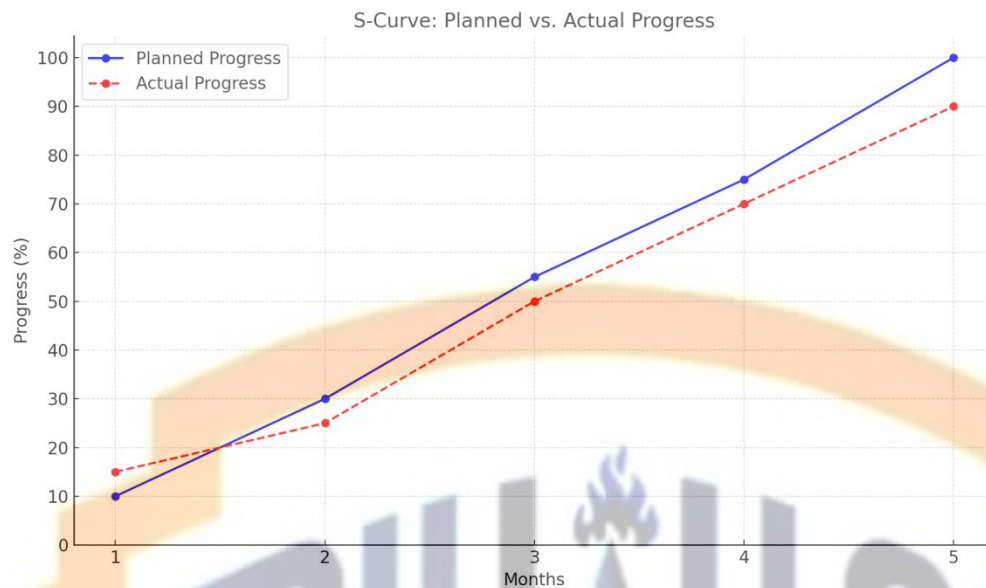
Steps for Drawing:

1. Plot the points for **Planned Progress** for each month:
(1, 10), (2, 30), (3, 55), (4, 75), and (5, 100).
 2. Plot the points for **Actual Progress** for each month:
(1, 15), (2, 25), (3, 50), (4, 70), and (5, 90).
 3. Connect the points for **Planned Progress** and **Actual Progress** with smooth curves.
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Step 3: Analysis of Gaps

1. **Month 1:** The actual progress is ahead of the planned progress by **+5%**.
Analysis: The project started ahead of schedule.
 2. **Months 2, 3, and 4:** The actual progress lags behind the planned progress by **-5%** each month.
Analysis: The project faced slight delays in these months.
 3. **Month 5:** The actual progress is behind the planned progress by **-10%**.
Analysis: The project completion is delayed significantly.
- **Early progress** in Month 1 provided an initial boost to the project.
 - From Month 2 onwards, **delays** occurred every month, with the gap widening by Month 5.
 - This indicates a **consistently delayed trend** starting from Month 2, requiring corrective action to bring the project back on track.
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Step 4: Creating the S-Curve



Question 2: Project Cost Performance Using S Curve

A project has a total budget of \$100,000. The planned cumulative expenditures are as follows:

- Month 1: \$10,000
- Month 2: \$30,000
- Month 3: \$60,000
- Month 4: \$90,000
- Month 5: \$100,000

The actual cumulative expenditures are as follows:

- Month 1: \$15,000
- Month 2: \$25,000
- Month 3: \$55,000
- Month 4: \$85,000
- Month 5: \$95,000

Question:

- (a) Plot the planned and actual expenditures on an S Curve.
- (b) Determine if the project is over-budget or under-budget.
- (c) What recommendations would you give to improve cost performance?

Solution:

Step 1: Tabular Representation of Data

Month	Planned Expenditure (\$)	Actual Expenditure (\$)	Difference (Actual - Planned)
1	\$10,000	\$15,000	+\$5,000
2	\$30,000	\$25,000	-\$5,000
3	\$60,000	\$55,000	-\$5,000
4	\$90,000	\$85,000	-\$5,000
5	\$100,000	\$95,000	-\$5,000

Step 2: Plotting the S-Curve

Explanation:

1. **Planned Expenditure** and **Actual Expenditure** data will be plotted on a graph (S-curve).
2. **X-axis:** Represents the months (1 to 5).
3. **Y-axis:** Represents cumulative expenditure in dollars (from \$0 to \$100,000).
4. Smooth curves will be drawn for planned and actual expenditures.

Step 3: Analysis of Cost Performance

Observations:

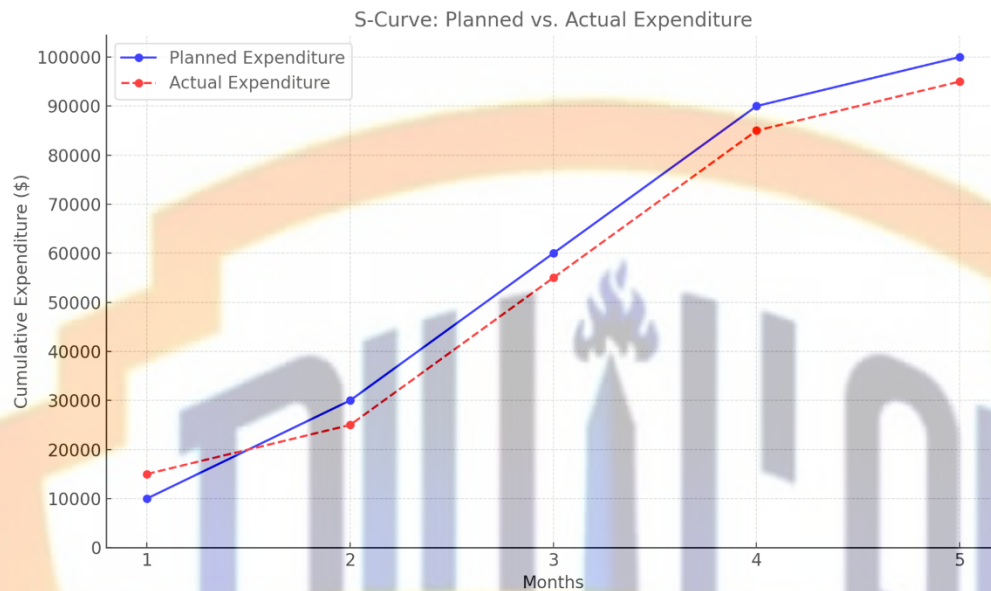
1. **Month 1:** The actual expenditure exceeds the planned expenditure by **+\$5,000**.
Analysis: Over-budget in the first month, possibly due to unplanned upfront costs.
2. **Months 2, 3, 4, and 5:** The actual expenditure is consistently **-\$5,000** less than the planned expenditure.
Analysis: The project is under-budget after Month 1.

Step 4: Recommendations

Based on the analysis:

1. **Address Variance in Initial Spending:**
 - The over-budget spending in Month 1 suggests inefficient planning or unexpected costs. Investigate the reasons and improve cost estimation for early phases.
2. **Monitor Future Spending:**
 - Consistent under-spending in later months might indicate delays in project execution or incomplete tasks. Ensure that under-spending does not compromise project quality.
3. **Budget Reallocation:**
 - If under-spending persists, assess whether savings can be reallocated to critical project activities to improve quality or meet deadlines.

Step 5: Plotting the S-Curve



Question 1: Critical Path Method (CPM)

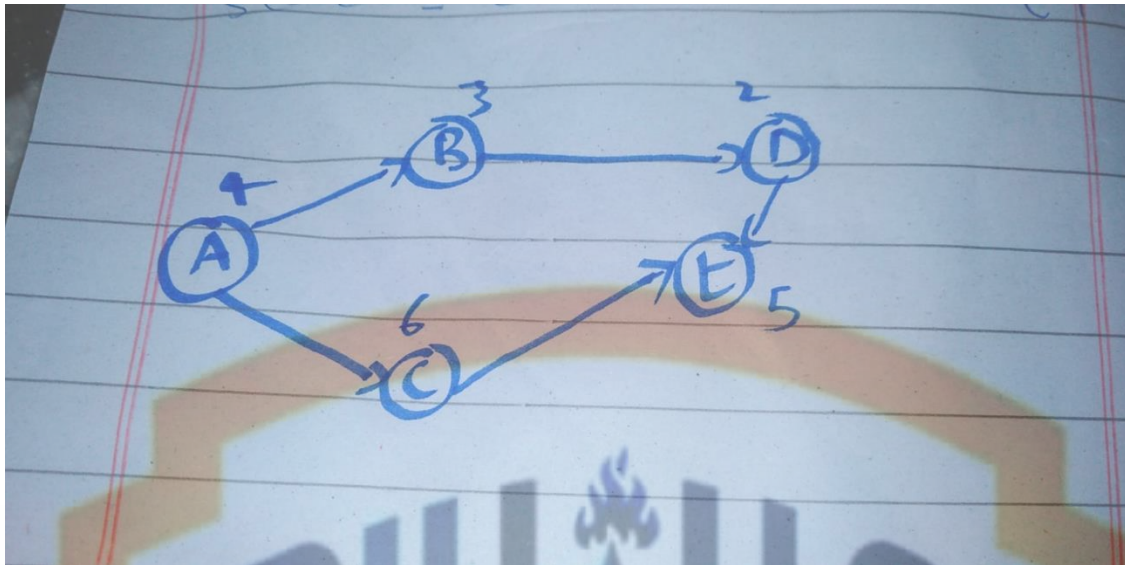
A project has the following activities and durations (in days):

Activity	Immediate Predecessor	Duration (Days)
A	-	4
B	A	3
C	A	6
D	B	2
E	C, D	5

Questions:

- Draw the project network using the Activity on Node (AON) method.
- Identify the critical path and its total duration.
- Calculate the slack for each activity.

Solution:



Step : Calculate
Forward Pass

ES

EF = ES + Duration

	ES	EF
A	0	$0 + 4 = 4$
B	4	$4 + 3 = 7$
C	4	$4 + 6 = 10$
D	7	$7 + 2 = 9$
E	$\max(10, 9) = 10$	$10 + 5 = 15$

Step = Calculate Backward pass

	LF	LS = LF - Duration
E	EF = 15	15 - 5 = 10
D	10	10 - 2 = 8
C	10	10 - 6 = 4
B	8	8 - 3 = 5
A	min(5, 4) = 4	4 - 4 = 0

Step 4: Calculate Slack

Activity	ES	EF	LS	LF	Slack
A	0	4	0	4	0
B	4	7	5	8	1
C	4	10	4	10	0
D	7	9	8	10	1
E	10	15	10	15	0

Step 5: Identify the Critical Path

The **Critical Path** is the sequence of activities with **Slack = 0**:

- A → C → E
- Total Duration = 4 (A) + 6 (C) + 5 (E) = 15 days

Question 2: Project Evaluation and Review Technique (PERT)

Activity	Immediate Predecessor	Optimistic Time (O)	Most Likely Time (M)	Pessimistic Time (P)
A	-	2	4	6
B	A	3	5	9
C	A	1	2	3
D	B, C	4	6	10

Questions:

- Calculate the expected time (T_e) for each activity using the formula:
 $T_e = (O + 4M + P) / 6$
- Draw the project network using Activity on Arrow (AOA).
- Determine the critical path based on the expected times.

Solution

(a) Calculate the Expected Time (T_e)

The formula to calculate the expected time for each activity is:

$$T_e = (O + 4M + P) / 6$$

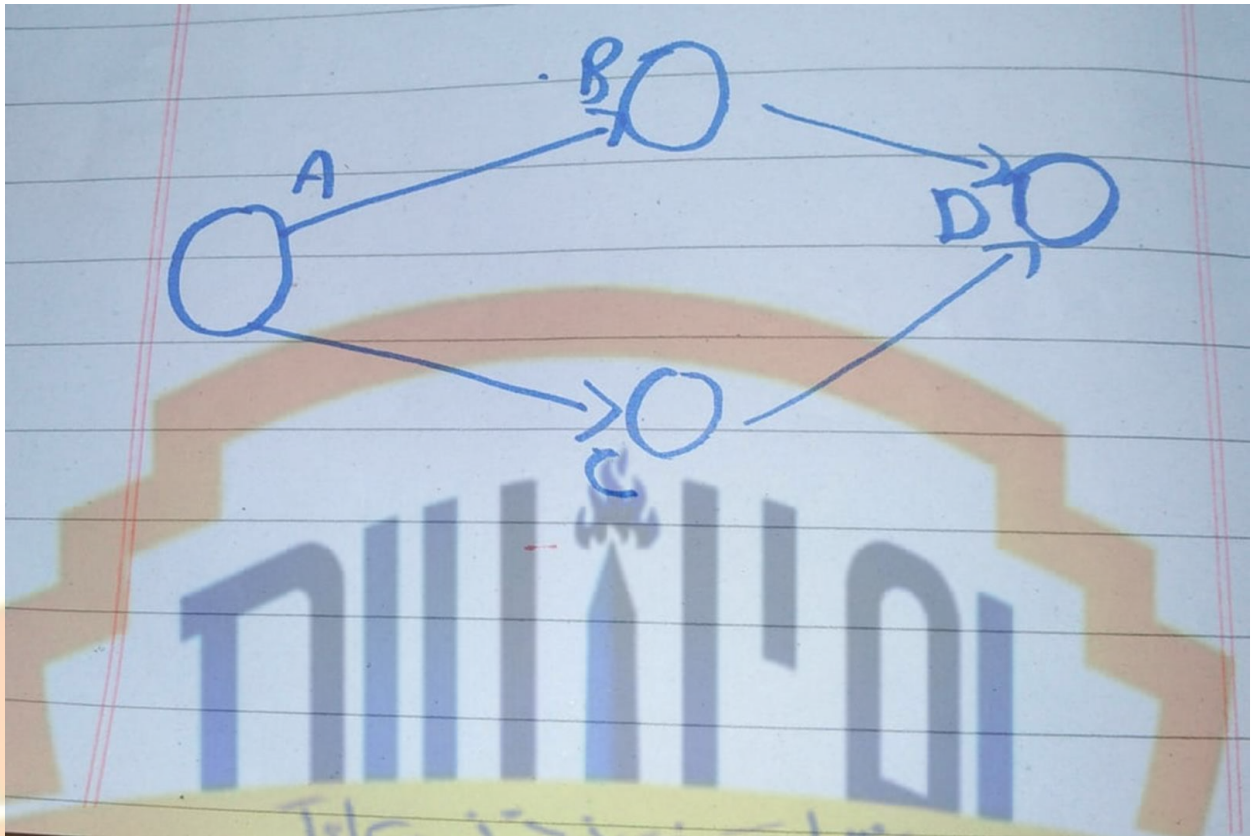
Given Data:

Activity	O (Optimistic Time)	M (Most Likely Time)	P (Pessimistic Time)
A	2	4	6
B	3	5	9
C	1	2	3
D	4	6	10

Expected Times:

Activity	Expected Time (T_e)
A	4
B	5.3
C	2
D	6.3

Network Diagram:



(c) Determine the Critical Path

To determine the critical path, we must calculate the total duration of each path and identify the longest one.

Paths in the Network:

1. **Path 1:** Start $\rightarrow$ A $\rightarrow$ B $\rightarrow$ D

$$\text{Duration} = \text{TeA} + \text{TeB} + \text{TeD} = 4 + 5.3 + 6.3 = 15.6$$

2. **Path 2:** Start $\rightarrow$ A $\rightarrow$ C $\rightarrow$ D

$$\text{Duration} = \text{TeA} + \text{TeC} + \text{TeD} = 4 + 2 + 6.3 = 12.3$$

Critical Path:

The **critical path** is the path with the longest duration:

Critical Path = A $\rightarrow$ B $\rightarrow$ D

Total Duration = 15.6 days

Final Results:

1. **Expected Times:**

- A = 4 days
- B = 5.3 days
- C = 2 days
- D = 6.3 days

2. **Critical Path:**

A→B→D

3. **Total Duration of Critical Path:**

15.6 days

